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VMM General Component Requirements - Continious power

GEEA 3.0

Vehicle Mode Management SWRS for ECU

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1 LIST OF REQUIREMENTS WITH DEVIATIONS

2 INTRODUCTION

2.1 What is the purpose of this document?

This document is a Vehicle Mode Management SWRS. It contains the software requirements for implementation of the Vehicle Mode Management Concept. This specification may contain generic requirements that are not applicable to specific ECUs.

2.2 Who should read this document?

This document is intended for:

- Developers of ECU software and hardware at the car OEM, ECU suppliers (Tier 1:s) or sub suppliers/contractors (Tier 2:s).
- Test engineers of electrical systems, ECUs or ECU software and hardware.

2.3 How do you receive more information?

More information can be found in the reference document lists. Questions and support issues may also be sent to the team within Geely responsible for a specific ECU or to the central Vehicle Mode Management team responsible for the specific project.

3 ECU SOFTWARE REQUIREMENTS

ID	Name	Version	Status	Last Changed
x04000000129C1CA3	VMM General Component Requirements	(6a1)	CS Released	2023-06-25

3.1 VMM General Component Requirements

ID	Name	Version	Status	Last Changed
x04000000129C1CA3	VMM General Component Requirements	(6a1)	CS Released	2023-06-25

Vehicle Mode Management related requirements affecting all electrical components.

3.1.1 CarMode

The Car Mode Manager (CMM) shall produce Car Mode states that reflect the life cycle states of the vehicle.

The main purposes of using Car Modes are:

- To achieve a common knowledge among vehicle nodes and vehicle functions about modes in the vehicle.
- To reduce current consumption in any mode that is not a "customer" mode in order to preserve the battery (power supply) quality throughout for instance factory and transport activities.
- To achieve a common mode in which specific functionality is wanted, e.g. during fuel certification where safe and repeatable tests are required.

Car Mode was originally designed to support manufacturing and logistics in realizing the functional requirements they had during factory and transport. It was then also possible to

make use of these modes to enable a low current consumption before final delivery to the customer, hence preserving the battery in a good as new state. Car Mode has then expanded to encompass other whole-vehicle-states that somehow puts the entire vehicle in a far from normal operational state, e.g. crash and Dyno mode. The Car Mode enables a certain vehicle set up which is not intended for the final customer, but which might improve development work, handling before final delivery and so on.

CarMode Signal Definition

The signal CarMod is an enum 3 bit signal that includes the modes mentioned below. The signal shall have protection against bit flip for safety classified transitions according to ISO 26262.

The signal can be combined with another signal, CarModSubTyp that defines substates to the CarMod signal. CarModSubTyp is an 3 bit integer signal with values according to the table below. CarModSubtyp enables expanding the signal in the future, without the need to make changes to all functions if only a few are affected by the change. All systems shall only look at the CarMod signal unless other is stated. Alternative: If a substate is received that is not defined, default CarModSubTyp shall be used (000).

The default value for CarMode is Normal.

CarMod encoding	3 bits	CarModSubTyp substate encoding	3 bits
Normal (CarModNorm)	000	Normal	000
		Factory Paused	001
		Factory Driving	010
		Transport Driving	011
		Future Use	100-111
Transport (CarModTrnsp)	001	Transport	000
		Future Use	001-111
Factory (CarModFcy)	010	Factory	000
		Future Use	001-111
Crash (CarModCrash)	011	Crash Level 1	000
		Crash Level 2	001
		Future Use	010-111

Dyno (CarModDyno)	101	Dyno2	000
		Dyno4	001
		Future Use	010-111

98022 v2 **Allowed CarModes in UsageMode Driving**

Purpose: It must be secured that CarMode is not set in receiver ECU unintentionally during driving. May cause function to turn off.
Source:
Verification Method:
Legacy ID: 211713

Systems must make sure that the correct CarMode is used by application. During UsageMode = Driving, only CarMode Normal, Crash and Dyno are valid.

CarMode is however not allowed to change to Dyno if UsageMode = Driving.

98026 v1 **Default value on CarMode**

Purpose: To avoid software upgrade and malfunction
Source:
Verification Method:
Legacy ID: 75269

The default value for CarMode (CarMod) subscribers shall be CarMode **NORMAL**.

Verification Method
Test and inspection

98029 v1 **Diagnostic trouble codes due to Car Mode Dyno**

Purpose: Not to generate diagnostic trouble code due to Car Mode Dyno.
Source:
Verification Method:
Legacy ID: 356602

Systems shall not go into reconfiguration (reset or similar) due to faults only related to dynamo testing environment.
If the test environment is the root cause for fault codes or signal quality degradation, this shall be neglected during dynamo mode as long as hardware is protected and safety is obtained.

3.1.2 UsageMode

The primary purpose of Usage Modes is to create a consistent behaviour when it comes to functional availability in the vehicle.

UsageMode will give a common understanding in the vehicle about user intention and will guide functions when it is allowed to operate (in combnation with a functional event trigger).
The primary function of the Usage Mode Manager (UMM) is to determine the access level of functionality in the vehicle based on the interpretation of user inputs. The level of access is described in modes and communicated to the rest of the vehicle via the signal UsgMod, generated by the UMM.

Usage Mode Signal Definition

The signal UsgModSts is a 4 bit enum signal that includes the modes mentioned above. The signal shall have protection against bit flip for safety classified transitions according to ISO 26262.

The default value for UsageMode is Inactive.

UsgModSts encoding	4 bits
Abandoned (UsgModAbdnd)	0000
Inactive (UsgModInActv)	0001
Convenience (UsgModCnvinc)	0010
Active (UsgModActv)	1011
Driving (UsgModDrvg)	1101

Table 3: Definitions of UsageMode states

98038 v1 **Default value on UsageMode**

Purpose: Try to not get stuck in an active state where the component consumes more current than necessary.
Source:
Verification Method:
Legacy ID: 211733

Default value on UsageMode shall be Inactive to not activate any functionality if not necessary.

An exception to this is if the FMEA or similar for the function finds that default value on UsageMode shall be other than UsageMode = Inactive.

98041 v1 **Limitations in Usage Mode Abandoned**

Purpose: To minimum current consumption and to avoid drained battery in a parked vehicle.
Source:
Verification Method:
Legacy ID: 266334

No functions/Systems or component is allowed to be active or keep the network continuously active in Usage Mode Abandoned.

If any function/system trigger is received in UM Abandoned either a request to change UM must be sent to VMM and UM has to be changed in order to allow the function to operate or this shall be approved by VMM in Electraical Architecture before it is implemented.

Only certain functions are allowed to request a UM change.

98042 v1 **Subscriber for Usage Mode**

Purpose: Correct functional behaviour.
Source:
Verification Method: Test
Legacy ID: 60488

A subscriber of the Usage Mode signal shall be able to handle any received Usage Mode signal without considering the previous received signal.

3.1.3 EEPM

The main purpose with Electrical Energy and Power Management (EEPM) is to control functional energy and power consumption based on the electrical energy and power availability, Usage Mode and Car Mode.

ElPowerLevel controls electrical loads depending on the electrical power availability in the vehicle. Typically depending on the electric power conditions. ElPowerLevel reduces loads to ensure voltage supply quality.

The ElPowerLevel contains the following Main states:

PwrLvlElecMai	PwrLvlElecSub	Description

(hexadecimal)	(hexadecimal)	
0	0-F	No additional limitations on electrical system consumption.
1	0-F	12V cold start (No support battery)
2	0-F	Engine standby
3	0-F	Increase power consumption
4	0-F	Active safety threat.
5	0-F	Fault in electrical system
6-F	0-F	Reserved for future use.

Table 4: ElPowerLevel Main states

ElEnergyLevel

The ElEnergylevel contains the following Sub states:

EgyLvlElecMain (hexadecimal)	EgyLvlElecSub (hexadecimal)	Description
0	0-F	Default
1	0	Critical electrical energy available

1	1	Very low electrical energy available, limit
1	2	Low electrical energy available, warning
1	3-F	Reserved for future use.
2	0-F	Reserved for future use.
3	0-F	Reserved for future use.
4	0-F	Electrical energy prewarning
5-F	0-F	Reserved for future use.

Table 5:ElEnergyLevel Main states

98052 v1 **Default value on ElEnergyLevel**

Purpose: When critical level of energy availability is reached elctrical loads shall be reduced/turned of to ensure vehicle start. When ElEnergyLevel 0 is sent there shall be no additional limitations on electrical power consumption.

Source:

Verification Method:

Legacy ID: 211754

The default value for ElEnergyLevel (EgyLvlElecMai) subscribers shall be 0 .

157715 v1 **Default value on ElPowerLevel**

Purpose: ElPowerLevel shall control 12V loads depending on the power precondition in the

vehicle. When ElPowerLevel 0 is sent there shall be no additional limitations on electrical power consumption.

Source:
Verification Method: Test
Legacy ID:

The default value for ElPowerLevel (PwrLvlElecMai) subscribers shall be 0.

3.1.4 Energy Consumption

Vehicle Mode Management requirements concerning minimising current consumption.

98057 v2 Current consumption, Go to SleepStandby in Abandoned, Inactive or Convenience mode

Purpose: All individual electric components have to be designed so the total electric consumption of the vehicle does not exceed the level that available electric power supply system can support. This is valid for all Usage Modes.
Source:
Verification Method: Test + Inspection
Legacy ID: 65343

When there is no communication, for example in UsageMode = Abandoned, Inactive or Convenience, all functions/systems and components/nodes shall go to Off / Sleep state within 15 sec after the network is released (i.e. no communication).

This requirement is valid when there is no safety requirement that keeps the function/component active. For example, this requirement is not applicable for safety reasons while the vehicle is still moving.

Functions that need more than 15 sec must apply for a deviation. The deviation must be approved by the VMM team at GEELY Electrical Architecture.

98059 v1 Current consumption, short activation time

Purpose: All individual electric components have to be designed so the total electric consumption of the vehicle does not exceed the level that available electric power supply system can support. This is valid for all Usage Modes.
Source:
Verification Method: Test
Legacy ID: 65341

Components shall not be active until the moment it is required to realise a function and shall be deactivated as soon as it is not required anymore.

For example if the customer activates a function the component that realises that function may then be activated, and shall be deactivated as soon as the customer deactivates the function or the function has been timed out.

Exception: During diagnostic communication session with the external tester activated the component can be active.

Verification Method
Test and inspection

98063 v2 Current consumption, No communication in Abandoned, Inactive or Convenience mode as default

Purpose: To limit energy consumption in the vehicle.
Source:
Verification Method: Test + Inspection
Legacy ID: 65342

Flexray, CAN and LIN networks shall by default be without signaling in UsageMode = Abandoned, Inactive or Convenience. A node shall never request communication unless requested by a function. It shall not activate or keep the network active only because it is power fed. Nor shall it request communication only because there is traffic on the bus.

When there is no communication in UsageMode = Abandoned, Inactive or Convenience, all nodes not carrying out or supporting a customer function shall go to Off/Sleep state and not consume more current than specified for this state.

98064 v3 Current consumption, max allowed for local modes

Purpose: Correct functional behaviour.
All individual functions have to be designed so the total electric consumption of the vehicle does not exceed the level that available electric power supply system can support.
Source:
Verification Method: Test + Inspection
Legacy ID: 60526

The max allowed current consumption per ECU/component is specified by VMM current consumption requirements（DPR）. There are typically requirements for local modes. Generally quiescent current in off/sleep below **0.1 mA** is required for a network connected component.

- Off /Sleep: The node has no activity and has reached its absolute lowest power consumption, but still need to be powered e.g. to maintain memory.
- Standby: The node has no functionality noticeable for the customer but is still active i.e. waiting to be activated by the customer or function.
- Active: The node has normal functionality i.e. activated and noticeable for the customer. Waiting for a button push before activating noticeable function is not considered to be an Active state, see Standby.

Measured data (and estimations) shall be presented for the node in the VMM section of the Load database and be updated to the latest, in a format that enables evaluation of requirement

values.

Note: DVM shall be used and reported back to VMM team at every release.

98065 v2 **No network activity due to polling in UsageMode Abandoned**

Purpose: Minimise current consumtion
Source:
Verification Method: Test + Inspection
Legacy ID: 142748

It is not allowed to keep alive or wake up other systems/components in UsageMode=Abandoned unless activation criterias are clearly specified in ECU application SWRS and agreed with VMM.

98072 v1 **Vehicle shut down procedure**

Purpose:
Source:
Verification Method:
Legacy ID:

All network communication shall during normal conditions stop within 10 sec after vehicle has entered UsageMode= Inactive.
In UsageMode=Inactive all components shall go to sleep within 15 sec after network communication has stopped.

Go to sleep mean to consume no more than quiescent current.

All deviations from this requirement must be agreed by VMM team in Electrical Arhictecture.

98076 v3 **Network/Component failsafe timeout**

Purpose: Remove conditions that can cause the component to stay awake "forever" and consume current.
Source:
Verification Method: Test + Inspection
Legacy ID: 142746

All ECU's must make sure that there exist no conditions that can keep the ECU to stay active "forever".
Conditions shall always contain a failsafe timeout that can shut down in case of error. This includes wakup/sleep as well as communication (VFC/PNC) criterias.

For exampel if an ECU is waiting for input before shut down, but is not receiving the signals within 120 sec, then the ECU and the network shall go to sleep.

Note: Examples of errors that can cause this:
- Implementation error

- Missing signal input
- Abnormal wakeup behaviour
- Timing error
- Button stuck
- Cyclic wake up or communication
- CAN bus Off
- synchronisation error on flexray networks
- Frame configuration error

Extensive testing must be done on all conditions for wake up/sleep and communication behaviour to secure no condition can cause ECU to get stuck and consume current.

98079 v2 **No continious network communication**

Purpose: Minimise the need for network communication in a parked vehicle.
Source:
Verification Method:
Legacy ID: 211835

All functions/systems shall be designed in a way that there is no need for continuous network communication when being in UsageMode Convenience, Inactive or Abandoned, to be able to carry out the functionality. Only short network communication with maximum 10 sec activation time is allowed.

This requirement is valid when there is no safety requirement that keeps the function/component active. For example, this requirement is not applicable for safety reasons while the vehicle is still moving.
Functions that need more than 10 sec must apply for a deviation. The deviation must be approved by the VMM team at GEELY Electrical Architecture before implementation.

98080 v2 **Current consumption, activation criterias**

Purpose: All individual electric components have to be designed so the total electric consumption of the vehicle does not exceed the level that available electric power supply system can support.
Source:
Verification Method:
Legacy ID: 211833

Components are not allowed to fully activate or stay active only due to communication on the networks in UsageMode Abandoned, Inactive and Convenience.

Components are not allowed to fully activate or stay active only due to being power fed (for example relay fed components) in UsageMode Abandoned, Inactive and Convenience.

If woken up by communication or power fed, the function/system must check on both UsageMode and the functional activation criteria before allowed to activate. If the functional activation criteria is not fulfilled the function /system shall not require the component to be active. Only parts of the component that are necessary for evaluating when to be active is allowed to be fed.

If not allowed to be active, the component shall enter a Standby mode and fulfill the Standby current consumption requirements.

3.1.5 Request and evaluation of Vehicle Mode Signals

Vehicle Mode Management requirements concerning requesting and evaluating VMM signals.

98101 v1 **Evaluation of VMM signals**

Purpose: To have a correct functional behaviour by evaluating CarMode, UsageMode, ElPowerLevel, ElEnergyLevel and make only one decision in the same processing period.
Source:
Verification Method:
Legacy ID: 75242

A subscriber of more than one VMM signal (CarMode, UsageMode, ElPowerLevel or ElEnergyLevel) shall have only one evaluation (taking all subscribed VMM signals into account) and take one decision leading to if the functions should be active or not.

The evaluation shall reflect the priority order that exist for the VMM signals by evaluating the signals in the following order:
- CarMode
- UsageMode
- ElPowerLevel and ElEnergyLevel

Verification method
Test and inspection

98102 v3 **Missing VMM signals**

Purpose:
Source:
Verification Method:
Legacy ID: 211654

If function requires valid Vehicle Modes (CarMode, UsageMode, ElPowerlevel or ElEnergyLevel) to activate or while active, but the signal is not received. The function shall by default be designed to be able to save or keep the last known value of Mode signales (recommended if functions have any kind of response time requirement).

If above is not possible, the function shall set the signals to the default value, see req "Default value on CarMode", "Default value on UsageMode", "Default value on ElPowerLevel" and "Default value on ElEnergyLevel."

NOTE! A change in Vehicle Mode signals will always be informed in the complete vehicle

157713 v1 VMM Global signal definition

Purpose: Distributing the global vehicle modes.

Source:
Verification Method: Analysis
Legacy ID:

The global Vehicle Mode Management signals are distributed to all subsystems with the signal group VehModMngtGlbSafe1.

When determining the availability of your subsystem from a complete vehicle perspective these signals shall be considered.

The Signal group contains the following Vehicle Mode management signals.

Signals in VehModMngtGlbSafe1	Description	How the signals affect your subsystem
UsgModSts	Provide UsageMode	Shall be Used by everyone.
CarModSts1	Provide CarMode	shall be used if needed.
CarModSubtypWdCarModSubtyp	Provide CarMode Sub type	Shall be used if additional input requirement from VMM exist.
EgyLvlElecMai	Provide main state of the ElEnergyLevel	Shall be used if additional input requirement from VMM exist.
EgyLvlElecSubtyp	Provide sub state of the ElEnergyLevel	Shall be used if additional input requirement from VMM exist.
PwrLvlElecMai	Provide main state of the ElPowerLevel	Shall be used if additional input requirement from VMM exist.
PwrLvlElecSubtyp	Provide sub state of the ElPowerLevel	Shall be used if additional input requirement from VMM exist.
FltEgyCnsWdSts	Provide current consumption fault detection	Shall be used if additional input requirement from VMM exist.

Verification method
Test and inspection